

What is a Tensor

Free Space and Tensor Product

We are allowed to perform additions and multiplications on vectors. A natural question is whether we can perform a multiplication of two vectors. It is, however, uninituitive to imagine the result of such multiplication. Therefore, we would first define the **free vector space**, by just *stapling* two vectors together as the result of the multiplications. Then, we will see how to make sense of the result.

Free Space

Let V and W be two vector spaces with the field F . The **free vector space** arising from the **Cartesian product** of V and W is a vector space taking **every distinct pair of vectors** from V and W as the basis. It is denoted by $F(V \times W)$.

A typical element of the free vector space has the following form:

$$T = \sum_{i=1}^n t_i(v_i, w_i),$$

given that $t_i \in F$, $v_i \in V$, and $w_i \in W$ for all i and that the pairs (v_i, w_i) are distinct.

Although the free vector space successfully defines the multiplication: the product of two vectors $v \in V$ and $w \in W$ is just the pair (v, w) , it does not have any nice properties, such as

$$(av, w) = a(v, w),$$

for $a \in F$, $v \in V$, and $w \in W$. But in the free vector space, if $av \neq v$, then $a(v, w)$ is not the same as (av, w) .

Bilinearity

Let's list out the desirable properties of the multiplication of two vectors:

- Some sort of distributivity:
 - $(v_1 + v_2, w) = (v_1, w) + (v_2, w)$
 - $(v, w_1 + w_2) = (v, w_1) + (v, w_2)$
- Property of scalar multiplication:
 - $(av, w) = a(v, w)$
 - $(v, aw) = a(v, w)$

Together, these properties are called **bilinearity**. In order to make these equalities hold, we take the **quotient space** of the free vector space by setting some elements to be 0.

Quotient Space and Subspace

To understand what is a **quotient space**, let's first start with a general vector space V and it's subset N . The quotient space of V by N , denoted by V/N , instead of containing the elements of V , contains the **cosets**. The coset of $v \in V$ is the set of all the elements of V that can be obtained by adding an element of N to v , denoted by $v + N$. In other words, the coset of v is the set

$$v + N = \{v + n \mid n \in N\}.$$

That is, V/N is the set of all $v + N$ for $v \in V$.

We also need to define the **equivalence relation** on V/N as follows: for $v_1 + N, v_2 + N \in V/N$, we say that $v_1 + N$ is equivalent to $v_2 + N$ if $v_1 - v_2 \in N$. In other words,

$$v_1 + N = v_2 + N \text{ if and only if } v_1 - v_2 \in N.$$

Now, one can show that if N is a **subspace** of V , then V/N is a vector space. To do so, we can define the addition and scalar multiplication of the quotient space as follows:

$$(v_1 + N) + (v_2 + N) = (v_1 + v_2) + N \text{ and } a(v + N) = (av) + N.$$

1. Well-definedness of addition and scalar multiplication:

Since we **redefined** the **equivalence relation** on V/N , we need to show that the addition and scalar multiplication of the quotient space are **well-defined**, that is, if $v_1 + N, v_2 + N, v'_1 + N, v'_2 + N \in V/N$ and $a \in F$ such that $v_1 + N = v'_1 + N$ and $v_2 + N = v'_2 + N$, then

$$(v_1 + v_2) + N = (v'_1 + v'_2) + N$$

and

$$(av) + N = (av') + N.$$

To show this, consider

$$(v_1 + v_2) - (v'_1 + v'_2) = (v_1 - v'_1) + (v_2 - v'_2).$$

Since N is a subspace, we have $v_1 - v'_1 \in N$ and $v_2 - v'_2 \in N$, which implies that $(v_1 + v_2) - (v'_1 + v'_2) \in N$. This shows that

$$(v_1 + v_2) + N = (v'_1 + v'_2) + N.$$

Similarly, consider

$$av - av' = a(v - v').$$

Since N is a subspace, we have $v - v' \in N$, which implies that $a(v - v') \in N$. This shows that

$$(av) + N = (av') + N.$$

Hence, the addition and scalar multiplication of the quotient space are well-defined.

2. Closure of addition and scalar multiplication:

We have to show that if $v_1 + N, v_2 + N \in V/N$ and $a \in F$, then

$$(v_1 + v_2) + N \in V/N$$

and

$$(av) + N \in V/N.$$

This is true because $v_1 + v_2 \in V$ and $av \in V$.

3. Associativity of addition:

We have to show that if $v_1 + N, v_2 + N, v_3 + N \in V/N$, then

$$v_1 + N + (v_2 + N + v_3 + N) = (v_1 + N + v_2 + N) + v_3 + N.$$

We can show that both sides are equal to $(v_1 + v_2 + v_3) + N$.

$$v_1 + N + (v_2 + N + v_3 + N) = v_1 + N + (v_2 + v_3) + N = (v_1 + v_2 + v_3) + N,$$

and

$$(V_1 + N + v_2 + N) + v_3 + N = (v_1 + v_2) + N + v_3 + N = (v_1 + v_2 + v_3) + N.$$

This shows that the addition of the quotient space is associative.

4. Commutativity of addition:

We have to show that if $v_1 + N, v_2 + N \in V/N$, then

$$v_1 + N + v_2 + N = v_2 + N + v_1 + N.$$

We can show that both sides are equal to $(v_1 + v_2) + N$.

$$v_1 + N + v_2 + N = (v_1 + v_2) + N,$$

and

$$v_2 + N + v_1 + N = (v_2 + v_1) + N = (v_1 + v_2) + N.$$

This shows that the addition of the quotient space is commutative.

5. Existence of additive identity:

We have to show that there exists an element $0_{V/N} \in V/N$ such that for any $v + N \in V/N$, we have

$$v + N + 0_{V/N} = v + N.$$

This is true because $0 + N$ is a coset in V/N and

$$v + N + 0 + N = v + (v + 0) + N = v + N.$$

6. Existence of additive inverse:

We have to show that for any $v + N \in V/N$, there exists an element $-(v + N) \in V/N$ such that

$$v + N + (-(v + N)) = 0_{V/N}.$$

This is true because $-v + N$ is a coset in V/N and

$$v + N + (-v) + N = (v - v) + N = 0 + N.$$

7. Compatibility of scalar multiplication with field multiplication:

We have to show that if $a, b \in F$ and $v + N \in V/N$, then

$$a(b(v + N)) = (ab)(v + N).$$

This is true because

$$a(b(v + N)) = a((bv) + N) = (abv) + N = (ab)(v + N).$$

8. Distributivity of scalar multiplication with respect to vector addition:

We have to show that if $a \in F$ and $v_1 + N, v_2 + N \in V/N$, then

$$a((v_1 + N) + (v_2 + N)) = a(v_1 + N) + a(v_2 + N).$$

This is true because

$$a((v_1 + N) + (v_2 + N)) = a((v_1 + v_2) + N) \\ = (a(v_1 + v_2)) + N = (av_1 + av_2) + N = a(v_1 + N) + a(v_2 + N).$$

9. Distributivity of scalar multiplication with respect to field addition:

We have to show that if $a, b \in F$ and $v + N \in V/N$, then

$$(a + b)(v + N) = a(v + N) + b(v + N).$$

This is true because

$$(a + b)(v + N) = ((a + b)v) + N = (av + bv) + N = a(v + N) + b(v + N).$$

Tensor Product

Now, let's return to the free vector space $F(V \times W)$. We want to make the bilinearity properties hold. To do so, let's consider the set all the elements of the free vector space in the following forms:

- $(v_1 + v_2, w) - (v_1, w) - (v_2, w)$
- $(v, w_1 + w_2) - (v, w_1) - (v, w_2)$
- $(av, w) - a(v, w)$
- $(v, aw) - a(v, w)$

and **their linear combinations**, with $v, v_1, v_2 \in V, w, w_1, w_2 \in W$, and $a \in F$. This set is a subspace of the free vector space, denoted by N . Then, we can take the quotient space of the free vector space by N , denoted by $F(V \times W)/N$.

We showed that the quotient space of a vector space by it's subspace is a vector space. Therefore, $F(V \times W)/N$ is a vector space, denoted by $V \otimes W$, called the **tensor product** of V and W . Instead of writing (v, w) , we write $v \otimes w$ to denote the product of $v \in V$ and $w \in W$.

Properties of Tensor Product

Decomposition of tensors

Let V and W be two vector spaces with the field F . Let $\{v_i\}_{i \in I}$ and $\{w_j\}_{j \in J}$ be two bases of V and W , respectively. Then, the set $\{v_i \otimes w_j\}_{i \in I, j \in J}$ is a basis of the tensor product $V \otimes W$.

To show this, consider an element $t \in F(V \times W)/N$, from the definition of the quotient space,

$$t = t' + N$$

for some $t' \in F(V \times W)$. Thus, we can decompose t' as follows:

$$t' = \sum_{i=1}^n t'_i(v_i, w_i),$$

where $t'_i \in F$, $v_i \in V$, and $w_i \in W$ for all i and that the pairs (v_i, w_i) are distinct. Therefore,

$$t = \sum_{i=1}^n t'_i(v_i, w_i) + N$$

From the definition of vector addition in the quotient space, we have

$$t = \sum_{i=1}^n t'_i((v_i, w_i) + N).$$

Then, since v_i and w_i can be decomposed as linear combinations of the basis vectors, we have

$$t = \sum_{i=1}^n t'_i \left(\left(\sum_{k \in I} a_{i,k} v_k \right) \otimes \left(\sum_{l \in J} b_{i,l} w_l \right) + N \right),$$

From the bilinearity properties, we have

$$t = \sum_{i=1}^n t'_i \left(\sum_{k \in I} \sum_{l \in J} a_{i,k} b_{i,l} (v_k, w_l) + N \right) \\ = \sum_{i=1}^n \sum_{k \in I} \sum_{l \in J} t'_i a_{i,k} b_{i,l} ((v_k, w_l) + N)$$

but since $v_k \otimes w_l$ is just $(v_k, w_l) + N$, we have

$$t = \sum_{i=1}^n \sum_{k \in I} \sum_{l \in J} t'_i a_{i,k} b_{i,l} v_k \otimes w_l.$$

This shows that any element of the tensor product can be decomposed as a linear combination of the basis vectors in $\{v_i \otimes w_j\}_{i \in I, j \in J}$.

In general, an element of the tensor product is written in the following form:

$$t = \sum_{i \in I} \sum_{j \in J} T^{ij} (v_i \otimes w_j).$$

Concrete Example of Tensor Product

Consider the vector space $\mathbb{R}^2$ with the standard basis $e_1 = (1, 0)$ and $e_2 = (0, 1)$. Then, the tensor product $\mathbb{R}^2 \otimes \mathbb{R}^2$ has the following basis:

$$e_1 \otimes e_1, e_1 \otimes e_2, e_2 \otimes e_1, e_2 \otimes e_2.$$

One can show that $\mathbb{R}^2 \otimes \mathbb{R}^2$ is isomorphic to $\mathbb{R}^4$ or a vector space of 2×2 real matrices.

Chained Tensor Products

Now, we have defined tensor as the product of two vectors, what about the product between a tensor and a vector or the product between two tensors?

Consider three vector spaces U, V , and W with the field F and the tensor product

$$(U \otimes V) \otimes W \text{ and } U \otimes (V \otimes W).$$

If $\{u_i\}_{i \in I}, \{v_j\}_{j \in J}$, and $\{w_k\}_{k \in K}$ are bases of U, V , and W , respectively, then the set $\{u_i \otimes v_j\}_{i \in I, j \in J}$ are the basis of $U \otimes V$. Therefore, a typical element of $(U \otimes V) \otimes W$ has the following form:

$$t = \sum_{i \in I} \sum_{j \in J} \sum_{k \in K} T^{ijk} ((u_i \otimes v_j) \otimes w_k).$$

Similarly, we can define the tensor product $U \otimes (V \otimes W)$ and a typical element of it has the following form:

$$s = \sum_{i \in I} \sum_{j \in J} \sum_{k \in K} S^{ijk} (u_i \otimes (v_j \otimes w_k)).$$

Consider the tensor product of $u \otimes v$ and w , where $u \in U, v \in V$, and $w \in W$. We have

$$(u \otimes v) \otimes w = \left(\sum_{i \in I} a_i u_i \otimes \sum_{j \in J} b_j v_j \right) \otimes \sum_{k \in K} c_k w_k = \sum_{i \in I} \sum_{j \in J} \sum_{k \in K} a_i b_j c_k ((u_i \otimes v_j) \otimes w_k)$$

and

$$u \otimes (v \otimes w) = \sum_{i \in I} a_i u_i \otimes \left(\sum_{j \in J} b_j v_j \otimes \sum_{k \in K} c_k w_k \right) = \sum_{i \in I} \sum_{j \in J} \sum_{k \in K} a_i b_j c_k (u_i \otimes (v_j \otimes w_k)).$$

This shows that these tensor products are **canonically isomorphic**, that is, there is a **natural** isomorphism between $(U \otimes V) \otimes W$ and $U \otimes (V \otimes W)$ such that the tensor product of u, v , and w is the same in both spaces. Therefore,

$$(U \otimes V) \otimes W \cong U \otimes (V \otimes W).$$

Multilinearity

Since $(U \otimes V) \otimes W$ is isomorphic to $U \otimes (V \otimes W)$, the intuition is that we can somehow remove the parentheses and write the tensor product as $U \otimes V \otimes W$. However, notice that as $(U \otimes V) \otimes W$ and $U \otimes (V \otimes W)$ produce different mathematical objects, $U \otimes V \otimes W$ does not have a well-defined meaning. Therefore, we cannot just write $U \otimes V \otimes W$ without any clarification. We need a proper definition for multiple tensor products.

Consider the vector spaces $V_1, \dots, V_n$ with the field F . We can define the **free vector space** arising from the Cartesian product of $V_1, \dots, V_n$, denoted by $F(V_1 \times \dots \times V_n)$, as the vector space taking every distinct n -tuple of vectors from $V_1, \dots, V_n$ as the basis.

Again we are interested in the following properties of the multiplication of n vectors:

- $(v_1, \dots, x + y, \dots, v_n) = (v_1, \dots, x, \dots, v_n) + (v_1, \dots, y, \dots, v_n)$
- $(v_1, \dots, kx, \dots, v_n) = k(v_1, \dots, x, \dots, v_n)$

where $v_i \in V_i$ for all i ; $x, y \in V_j$ for some j , and $k \in F$. The above properties are called **multilinearity**. With the same idea as for the tensor product of two vector spaces, we can take the quotient space of the free vector space by the subspace N generated by all the elements in the above forms and their linear combinations. The resulting quotient space is called the **tensor product** of $V_1, \dots, V_n$, denoted by $V_1 \otimes \dots \otimes V_n$ or

$$\bigotimes_{i=1}^n V_i.$$

Since $F(V_1 \times \dots \times V_n)/N$ is a quotient space of a vector space by its subspace, it is a vector space.

In the same way as the tensor product of two vector spaces, we can show that a typical element of the tensor product of n vector spaces has the following form:

$$t = \sum_{i_1 \in I_1} \sum_{i_2 \in I_2} \dots \sum_{i_n \in I_n} T^{i_1 i_2 \dots i_n} (v_{i_1}^1 \otimes v_{i_2}^2 \otimes \dots \otimes v_{i_n}^n),$$

where $\{v_{i_j}^j\}_{i_j \in I_j}$ is a basis of V_j for all j . That is, the set

$$\{v_{i_1}^1 \otimes v_{i_2}^2 \otimes \dots \otimes v_{i_n}^n\}_{i_1 \in I_1, i_2 \in I_2, \dots, i_n \in I_n}$$

is a basis of the tensor product of n vector spaces.

Now, consider $u \otimes v \otimes w$, where $u \in U, v \in V$, and $w \in W$. We have

$$u \otimes v \otimes w = \sum_{i \in I} a_i u_i \otimes \sum_{j \in J} b_j v_j \otimes \sum_{k \in K} c_k w_k = \sum_{i \in I} \sum_{j \in J} \sum_{k \in K} a_i b_j c_k (u_i \otimes v_j \otimes w_k).$$

This shows that all three different tensor products, $(U \otimes V) \otimes W$, $U \otimes (V \otimes W)$, and $U \otimes V \otimes W$ are all **canonically isomorphic**, that is,

$$U \otimes V \otimes W \cong (U \otimes V) \otimes W \cong U \otimes (V \otimes W).$$

Moreover, as the result of the tensor product of u, v , and w generates the same coordinates for corresponding basis vectors in all three spaces, the **particular result** of the tensor product of u, v , and w is the same in all three spaces. Therefore, we can write

$$u \otimes v \otimes w \cong (u \otimes v) \otimes w \cong u \otimes (v \otimes w).$$

Thus, any operations performed between different tensor products of u, v , and w can be performed by first transforming the tensors through the isomorphic mapping into the same space and then performing the operations.

For instance, if we want to add elements between $(U \otimes V) \otimes W$ and $U \otimes (V \otimes W)$, we can first transform the elements in $(U \otimes V) \otimes W$ into $U \otimes (V \otimes W)$ through the isomorphic mapping and then perform the addition.

Universal Property of Tensor Product

The **universal property** of the tensor product states that for any vector space X and any **multilinear map** $f : V_1 \times \dots \times V_n \rightarrow X$, there exists a **unique linear map** $g : V_1 \otimes \dots \otimes V_n \rightarrow X$ such that

$$f = g \circ \varphi,$$

where $\varphi : V_1 \times \dots \times V_n \rightarrow V_1 \otimes \dots \otimes V_n$ defined by $\varphi(v_1, \dots, v_n) = v_1 \otimes \dots \otimes v_n$ for all $v_i \in V_i$ for all i . To prove this, consider the free vector space $F(V_1 \times \dots \times V_n)$. Since f is a multilinear map, we can extend f to a linear map $f' : F(V_1 \times \dots \times V_n) \rightarrow X$ by defining it as follows:

$$\tilde{f} \left(\sum_{k=1}^p t_k (v_k^1 \otimes v_k^2 \otimes \dots \otimes v_k^n) \right) = \sum_{k=1}^p t_k f(v_k^1, v_k^2, \dots, v_k^n),$$

where $\{v_k^j\}_{j \in J}$ for all j and $t_k \in F$ for all k .

- Let's first show that if $n \in N$, the subspace generated in the process of building the tensor product, then

$$\tilde{f}(n) = 0.$$

From the definition of N , we know that n is a linear combination of the following elements:

- $(v_1, \dots, x + y, \dots, v_n) - (v_1, \dots, x, \dots, v_n) - (v_1, \dots, y, \dots, v_n)$
- $(v_1, \dots, kx, \dots, v_n) - k(v_1, \dots, x, \dots, v_n)$

where $v_i \in V_i$ for all i ; $x, y \in V_j$ for some j , and $k \in F$. Let's represent those elements as e_i , then n can be written as

$$n = \sum_{i=1}^m n_i e_i.$$

Hence, we have

$$\tilde{f}(n) = \tilde{f} \left(\sum_{i=1}^m n_i e_i \right) = \sum_{i=1}^m n_i \tilde{f}(e_i).$$